Mosaics of resistance: Archaeal contributions to ARG dissemination across ecosystems and domains

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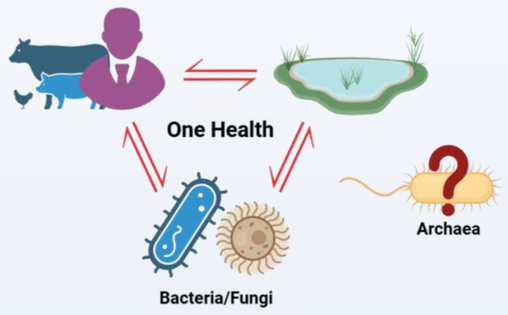
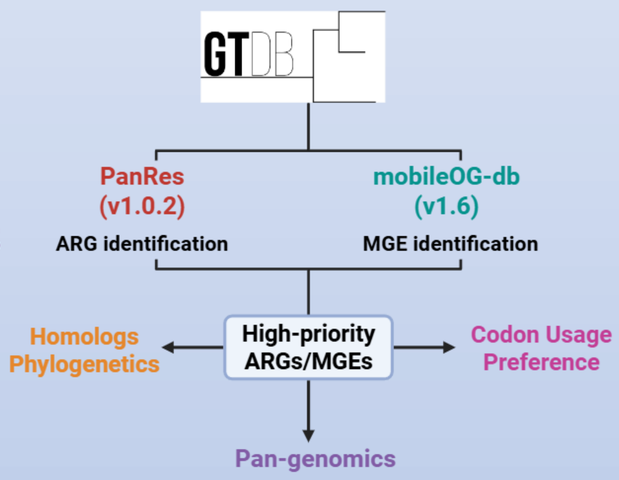
Introduction



- **The One Health Blind Spot:** Antimicrobial Resistance (AMR) is a critical threat under the One Health framework, while **archaea** remain an overlooked reservoir despite their ubiquity.
- **Ecological Context:** Archaea co-exist with resistant bacteria in guts/wastewater but also thrive in extreme environments where antibiotics degrade.
- **Key Question:**
  - ➢ Do archaea harbour clinically relevant antimicrobial resistance genes **(ARGs)**?
  - ➢ Can they exchange ARGs with bacteria via horizontal gene transfer **(HGT)**?
  - ➢ Who is the **origin**?

Methods



- **Data Collection:** Screened 12,477 archaeal genomes (GTDB v220), including metagenome-assembled genomes (MAGs), pure culture genomes (PCGs) and single-cell genomes (SCGs).
- **ARG Screening:** Aligned against PanRes database (Identity ≥70%, Coverage ≥60%).
- **MGE Association:** Identified mobile elements using mobileOG-db; filtered for proximity (<10kb) to ARGs.
- **Origin Tracing:** Multi-validation framework:
  - ➢ Homologs and Phylogenetics (MAFFT, RAxML)
  - ➢ Pan-genome Analysis (Roary)
  - ➢ Codon Usage Preference (RSCU/MILC/GC3)

Conclusion

- **Ecological Bridge:** Archaea serve as critical reservoirs, preserving ARGs in extreme niches and facilitating cross-ecosystem dissemination (mainly **Bacteria → Archaea** Flow).
- **Unique AMR Systems:** Extremophiles exhibit intrinsic resistance patterns distinct from bacteria, suggesting undiscovered archaeal-specific mechanisms.

Future Directions

- Develop archaea-specific ARG databases.
- Experimental validation of transfer rates in co-culture systems.

Results

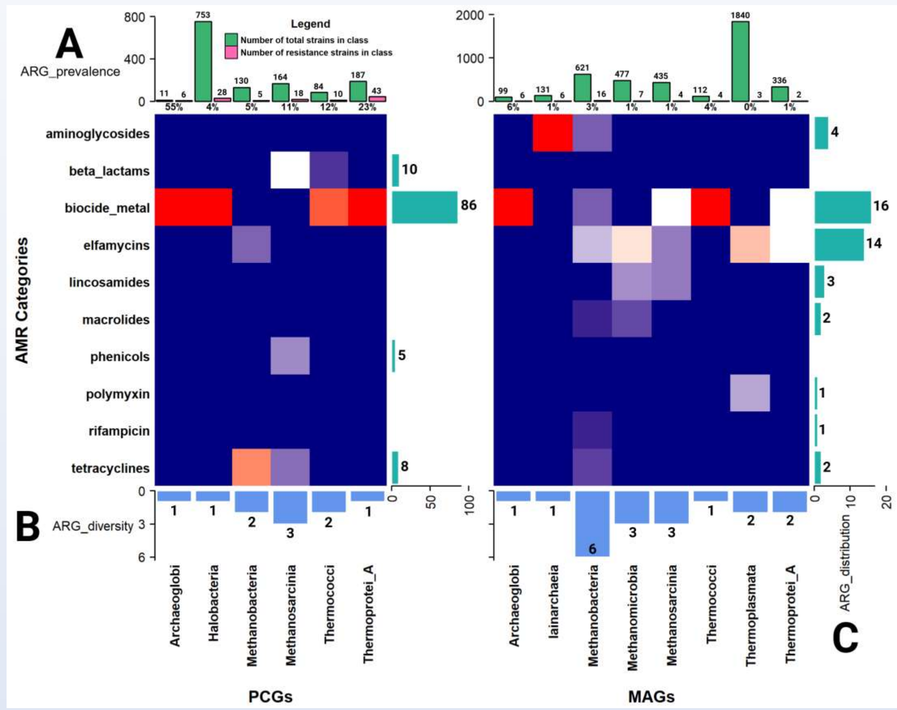


Figure 1. Prevalence, diversity and distribution of the ARGs in archaea between the PCGs and MAGs.

- **Host-Associated Methanogens (*Methanobacteria*, *Methanosarcinia*):**
  - ➢ Rich in Antibiotic ARGs (Tetracycline, Beta-lactams).
  - ➢ Co-localised with ARB in guts/sludge → Suggests HGT.
- **Extremophiles (*Halobacteria*, *Thermococci*):**
  - ➢ Dominated by Biocide/Metal ARGs (Efflux pumps).
  - ➢ Rare antibiotic ARGs → Likely Intrinsic/Vertical Evolution.
- **Prevalence:** <5% of genomes carry ARGs, but diversity is high in anthropogenic niches.

Bacteria → Archaea
Global Spread HGT?

Archaeal Resistome Dichotomy

Antibiotic ARGs is a borrowed trait from **bacteria (HGT)**, while **metal/biocide** resistance is an ancient, **intrinsic adaptation** to extreme niches.

Archaea → Archaea
Endogenous Mechanism?

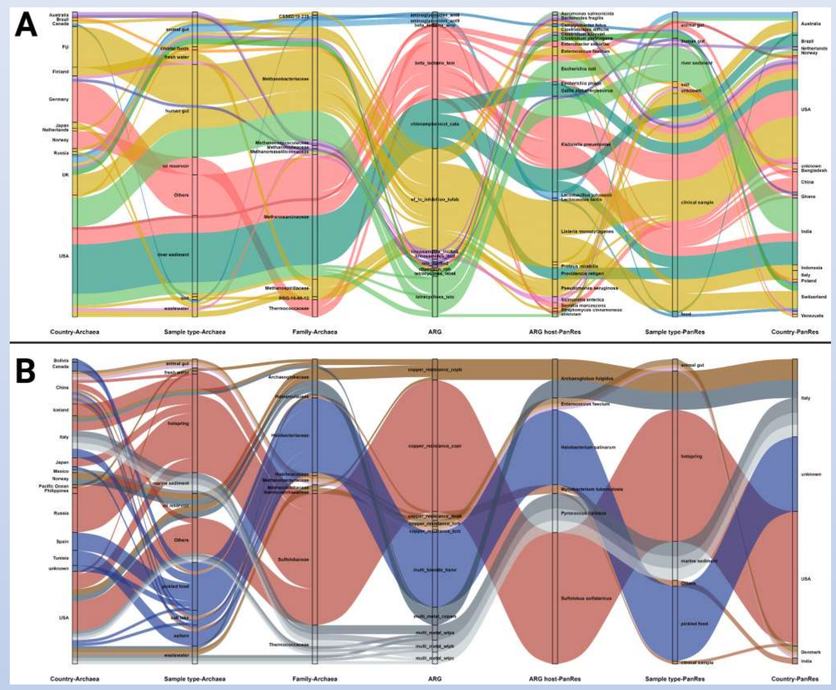
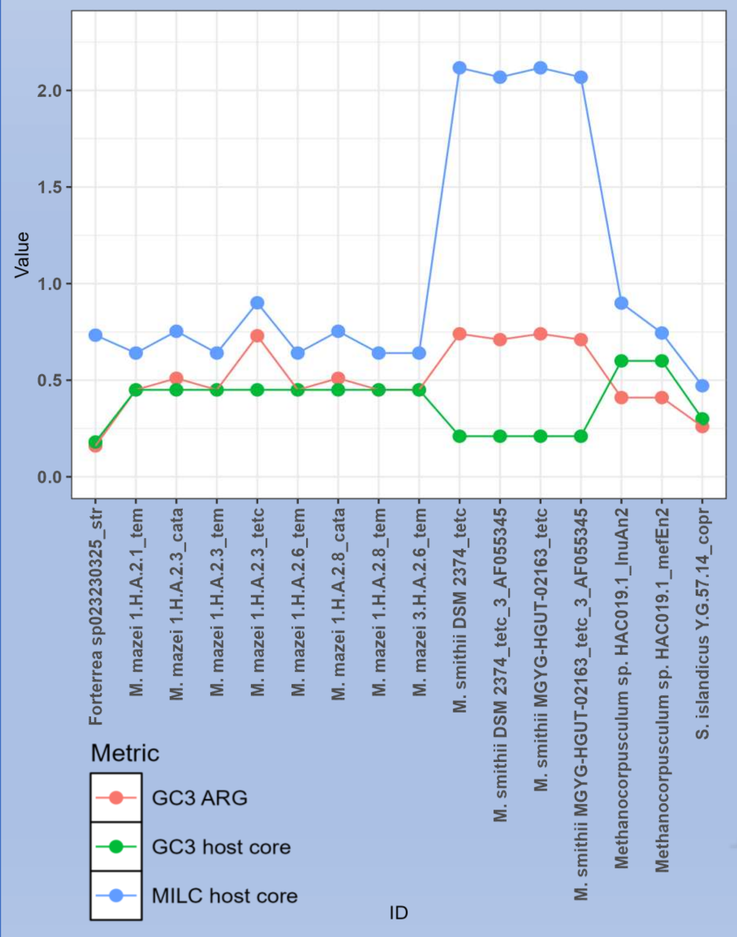


Figure 2. Metadata relationships between archaeal strains harbouring ARGs and the original microbial hosts of these ARGs.



- **Antibiotic ARGs:** Most ARGs (*tetc, lnuAn2, mefEn2*) exhibit **MILC values significantly deviating from 0.5** (Blue lines), indicating their codon usage differs markedly from their archaeal hosts. Concurrently, these genes show significant divergence in GC3 content compared to their host genomes. This dual discrepancy strongly supports a bacterial origin via HGT.
- **Endogenous ARGs:** In contrast, the *copr* gene in *S. islandicus* displays a MILC value close to 0.5 and a GC3 content nearly identical to its host. This concordance confirms *copr* as an endogenous core gene, vertically inherited within the *Sulfolobus* lineage.

Figure 3. Codon usage preference and GC3 content divergence of high-priority transferable ARGs and their archaeal host.
The MILC index measures CUP divergence, normalised such that 0.5 indicates perfect concordance with the reference genome (suggesting endogenous origin), while values > 0.5 indicate significant divergence (suggesting horizontal acquisition).

References

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